
Name : Abhivarsha
Roll No : 261100690029

Compare permissions on Regular Files and Directories

- Create a regular file named sample.txt and a directory named sample_dir. Create at least one file inside sample_dir.
- Apply each of the following permissions using chmod. Perform the specified operations and record your observations.

Permission	Regular File (sample.txt) – Test	Observation	Directory (sample_dir) – Test	Observation
400 (r--)	Try to read and modify the file	Read succeeds; modification fails.	Try to list and enter the directory	Listing succeeds; entering succeeds if directory has r but not x? Correction> listing succeeds, entering fails because x is required.
200 (-w-)	Try to read and modify the file	Read fails; writing/modification succeeds (subject to how the program opens the file).	Try to list, enter, and create a file	Listing fails; entering fails; creating files fails because x is required to access directory entries.
100 (--x)	Try to read and execute the file	Reading fails; execution may succeed only if the file is a valid executable/script and the required interpreter/access conditions are met.	Try to list and enter the directory	Listing fails; entering succeeds.
500 (r-x)	Try to read and modify the file	Reading succeeds; modification fails.	Try to list and enter the directory	Listing succeeds; entering succeeds.
600 (rw-)	Try to read and modify the file	Reading and modification succeed.	Try to list, enter, and create a file	Listing succeeds; entering fails; creating a file fails.
700 (rwx)	Try to read, modify,	Reading and modification succeed; execution is allowed by the permission bits (assuming the	Try to list, enter, create, and delete files	All operations succeed.

	and execute the file	contents are executable).		
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Create a regular file named sample.txt and a directory named sample_dir. Create at least one file inside sample_dir.

Apply each of the following permissions using chmod. Perform the specified operations and record your observations.

400(r - -)

```
kali@kali: ~  
Session Actions Edit View Help  
  
(kali@kali)-[~]  
$ touch sample.txt  
  
(kali@kali)-[~]  
$ mkdir sample_dir  
  
(kali@kali)-[~]  
$ touch sample_dir/file1.txt  
  
(kali@kali)-[~]  
$ ls -l  
total 52  
drwxr-xr-x 2 kali kali 4096 Aug 8 11:43 Desktop  
drwxr-xr-x 2 kali kali 4096 Aug 15 11:08 Documents  
drwxr-xr-x 2 kali kali 4096 Aug 15 11:03 Downloads  
drwxrwxr-x 2 kali kali 4096 Aug 10 12:30 example  
drwxrwxr-x 3 kali kali 4096 Aug 10 15:11 linuxOS_2026  
drwxr-xr-x 2 kali kali 4096 Aug 8 11:43 Music  
drwxrwxr-x 3 kali kali 4096 Aug 11 07:05 phishing_simulator  
drwxr-xr-x 2 kali kali 4096 Aug 8 11:43 Pictures  
drwxr-xr-x 2 kali kali 4096 Aug 8 11:43 Projects  
drwxr-xr-x 2 kali kali 4096 Aug 8 11:43 Public  
drwxrwxr-x 2 kali kali 4096 Aug 15 11:17 sample_dir  
-rw-rw-r-- 1 kali kali 0 Aug 15 11:17 sample.txt  
drwxr-xr-x 2 kali kali 4096 Aug 8 11:43 Templates  
drwxr-xr-x 2 kali kali 4096 Aug 8 11:43 Videos
```

Chmod 400 simple.txt

```
(kali@kali)-[~]  
$ chmod 400 sample.txt  
  
(kali@kali)-[~]  
$ chmod 400 sample_dir  
  
(kali@kali)-[~]  
$ ls -ld sample.txt sample_dir  
dr----- 2 kali kali 4096 Aug 15 11:17 sample_dir  
-r----- 1 kali kali 0 Aug 15 11:17 sample.txt
```

Chmod 400 sample_dir

```
(kali@kali)-[~]  
$ ls sample_dir  
ls: cannot access 'sample_dir/file1.txt': Permission denied  
file1.txt  
  
(kali@kali)-[~]  
$ cd sample_dir  
cd: permission denied: sample_dir
```

200(-w-)

```
(kali㉿kali)-[~]
$ chmod 200 sample.txt

(kali㉿kali)-[~]
$ chmod 200 sample_dir

(kali㉿kali)-[~]
$ ls -ld sample.txt sample_dir
d-w----- 2 kali kali 4096 Aug 15 11:17 sample_dir
--w----- 1 kali kali    0 Aug 15 11:17 sample.txt

(kali㉿kali)-[~]
$
```

Chmod 200 sample.txt

```
(kali㉿kali)-[~]
$ cat sample.txt
cat: sample.txt: Permission denied

(kali㉿kali)-[~]
$ echo "Hello" > sample.txt

(kali㉿kali)-[~]
$
```

Chmod 200 sample_dir

```
(kali㉿kali)-[~]
$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied

(kali㉿kali)-[~]
$ cd sample_dir
cd: permission denied: sample_dir

(kali㉿kali)-[~]
$ touch sample_dir/file2.txt
touch: cannot touch 'sample_dir/file2.txt': Permission denied

(kali㉿kali)-[~]
$
```

100(--x)

```
(kali㉿kali)-[~]
$ chmod 100 sample.txt

(kali㉿kali)-[~]
$ chmod 100 sample_dir

(kali㉿kali)-[~]
$ ls -ld sample.txt sample_dir
d--x----- 2 kali kali 4096 Aug 15 11:17 sample_dir
--x----- 1 kali kali    6 Aug 15 12:32 sample.txt

(kali㉿kali)-[~]
$
```

Chmod 100 sample.txt

```
(kali㉿kali)-[~]  
$ cat sample.txt  
cat: sample.txt: Permission denied  
  
(kali㉿kali)-[~]  
$ ./sample.txt  
zsh: permission denied: ./sample.txt  
  
(kali㉿kali)-[~]  
$
```

Chmod 100 sample_dir

```
(kali㉿kali)-[~]  
$ ls sample_dir  
ls: cannot open directory 'sample_dir': Permission denied  
  
(kali㉿kali)-[~]  
$ cd sample_dir  
  
(kali㉿kali)-[~/sample_dir]  
$ pwd  
/home/kali/sample_dir
```

500(r-x)

```
(kali㉿kali)-[~]  
$ chmod 500 sample.txt  
  
(kali㉿kali)-[~]  
$ chmod 500 sample_dir  
  
(kali㉿kali)-[~]  
$ ls -ld sample.txt sample_dir  
dr-x----- 2 kali kali 4096 Aug 15 11:17 sample_dir  
-r-x----- 1 kali kali   6 Aug 15 12:32 sample.txt  
  
(kali㉿kali)-[~]  
$
```

Chmod 500 sample.txt

```
(kali㉿kali)-[~]  
$ cat sample.txt  
Hello  
  
(kali㉿kali)-[~]  
$ echo "Hi" > sample.txt  
zsh: permission denied: sample.txt
```

Chmod 500 sample_dir

```
(kali㉿kali)-[~]  
$ ls sample_dir  
file1.txt  
  
(kali㉿kali)-[~]  
$ cd sample_dir  
  
(kali㉿kali)-[~/sample_dir]  
$ pwd  
/home/kali/sample_dir  
  
(kali㉿kali)-[~/sample_dir]  
$
```

600(rw-)

```
(kali㉿kali)-[~]  
$ chmod 600 sample.txt  
  
(kali㉿kali)-[~]  
$ chmod 600 sample_dir  
  
(kali㉿kali)-[~]  
$ ls -ld sample.txt sample_dir  
drw----- 2 kali kali 4096 Aug 15 11:17 sample_dir  
-rw----- 1 kali kali    6 Aug 15 12:32 sample.txt
```

Chmod 600 sample.txt

```
(kali㉿kali)-[~]  
$ cat sample.txt  
Hello  
  
(kali㉿kali)-[~]  
$ echo "Goodmorning" > sample.txt  
  
(kali㉿kali)-[~]  
$ cat sample.txt  
Goodmorning
```

700(rwx)

```
(kali㉿kali)-[~]  
$ chmod 700 sample.txt  
  
(kali㉿kali)-[~]  
$ chmod 700 sample_dir  
  
(kali㉿kali)-[~]  
$ ls -ld sample.txt sample_dir  
drwx----- 2 kali kali 4096 Aug 15 11:17 sample_dir  
-rwx----- 1 kali kali    12 Aug 15 12:44 sample.txt
```


Chmod 700 sample.txt

```
(kali㉿kali)-[~]
└─$ cat sample.txt
Goodmorning

(kali㉿kali)-[~]
└─$ echo "Goodnight" > sample.txt

(kali㉿kali)-[~]
└─$ cat sample.txt
Goodnight

(kali㉿kali)-[~]
└─$ ./sample.txt
./sample.txt: 1: Goodnight: not found
```

Chmod 700 sample_dir

```
(kali㉿kali)-[~]
└─$ cat sample.txt
Goodnight

(kali㉿kali)-[~]
└─$ ./sample.txt
./sample.txt: 1: Goodnight: not found

(kali㉿kali)-[~]
└─$ cd sample_dir

(kali㉿kali)-[~/sample_dir]
└─$ touch new.txt

(kali㉿kali)-[~/sample_dir]
└─$ ls
file1.txt  new.txt

(kali㉿kali)-[~/sample_dir]
└─$ rm new.txt

(kali㉿kali)-[~/sample_dir]
└─$ ls
file1.txt
```

Chmod 700 sample_dir

```
(kali㉿kali)-[~]
└─$ ls sample_dir
ls: cannot access 'sample_dir/file1.txt': Permission denied
file1.txt

(kali㉿kali)-[~]
└─$ cd sample_dir
cd: permission denied: sample_dir

(kali㉿kali)-[~]
└─$ touch sample_dir/file3.txt
touch: cannot touch 'sample_dir/file3.txt': Permission denied
```

Questions:

Q1. What does read (r) permission allow you to do with a regular file?

> It allows you to open and view the actual contents of the file

Q2. What does read (r) permission allow you to do with a directory?

> It allows you to read the directory's list of contents. You can see the names of the files and folders inside it

Q3. What does write (w) permission mean for a regular file?

> It allows you to write, modify or overwrite the contents of the file.

Q4. What does write (w) permission mean for a directory?

> It allows you to alter the directory's contents create new files, delete existing files, or rename files within that directory.

Q5. What does execute (x) permission mean for a regular file?

> It allows the operating system to run the file as a program or script .

Q6. Why is execute (x) permission important for accessing a directory?

> You need execute permission to enter the directory , access its metadata, or access any of the files inside it.

Q7. Which permission combination allows the owner to fully access a regular file?

> 7(rwx) or 700 for owner-only full access.

Q8. Based on your observations, explain why the same r, w, and x permissions behave differently for a regular file and a directory.

i. File >

When you apply permissions to a file, you are directly controlling what someone can do to the information inside it. Read permission means you are allowed to open the document and read the content. Write permission means you are allowed to change, add, or modify document. Execute permission means that if the document is a set of instructions or a program, you are allowed to actually run those files.

ii. Directory >

Read permission allows you to look at the directory's list, see the names of all the files in it. Write permission gives you the authority to alter that directory, create new files, remove existing files etc.. Execute permission acts as the key to get inside the directory. Even if we know exactly what is inside the directory, without execute permission, the directory is locked and you cannot use any of the files inside.
